1. (1) $a_n = \frac{3+5n^2}{n+n^2} = \frac{(3+5n^2)/n^2}{(n+n^2)/n^2} = \frac{5+\frac{3}{n^2}}{1+\frac{1}{n}}$

So $a_n \to \frac{5+0}{1+0} = 5$ as $n \to \infty$. Converges.

(2) $a_n = \frac{3+5n^2}{1+n} = \frac{(3+5n^2)/n}{(1+n)/n} = \frac{\frac{3}{n}+5n}{\frac{1}{n}+1}$

So $a_n \to \infty$ as $n \to \infty$ diverges.

(3) $a_n = \frac{n^4}{n^3+2n} = \frac{n^4/n^3}{(n^3+2n)/n^3} = \frac{n}{1-\frac{2}{n^2}}$

So $a_n \to \infty$ as $n \to \infty$ diverges.

(4) $a_n = \sqrt{\frac{1+4n^2}{1+n^2}} = \sqrt{\frac{(1+4n^2)/n^2}{(1+n^2)/n^2}} = \sqrt{\frac{\frac{1}{n^2}+4}{\frac{1}{n^2}+1}} \to \sqrt{4} = 2$ as $n \to \infty$ converges.

(5) $a_n = \frac{n^2}{\sqrt{n^4+4n}} = \frac{n^2/\sqrt{n^3}}{\sqrt{n^4+4n}/\sqrt{n^3}} = \frac{\sqrt{n}}{\sqrt{1+\frac{4}{n^3}}}$

So $a_n \to \infty$ as $n \to \infty$ diverges.

2. (1) $a_n = 2 + (0.86)^n \to 2+0 = 2$ as $n \to \infty$ Since $\lim_{n\to\infty} (0.86)^n = 0$ converges

(2) $a_n = 3^n 7^{-n} = (\frac{3}{7})^n$. So $\lim_{n\to\infty} a_n = 0$ converges

(3) $a_n = \frac{4^n}{1+9^n} = \frac{4^n/9^n}{(1+9^n)/9^n} = \frac{(\frac{4}{9})^n}{(\frac{1}{9})^n+1} \to \frac{0}{0+1} = 0$ as $n \to \infty$. converges.

(4) $a_n = \cos(\frac{n\pi}{n+1}) = \cos\left(\frac{n\pi/n}{(n+1)/n}\right) = \cos(\frac{\pi}{1+\frac{1}{n}})$

So $a_n \to \cos\pi = -1$ as $n \to \infty$. Converges.

(5) $a_n = \sin n$. This sequence diverges since the terms don't approach any particular real number as $n \to \infty$

The terms take on values between -1 and 1. diverges.

3. (1) $a_n = \ln(n+1) - \ln n = \ln(\frac{n+1}{n}) = \ln(1+\frac{1}{n}) \to \ln 1 = 0$ as $n \to \infty$. Converges.

(2) $a_n = n\sin(\frac{1}{n}) = \frac{\sin(\frac{1}{n})}{\frac{1}{n}}$

$\lim_{x\to\infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}} = \lim_{t\to 0} \frac{\sin t}{t}$ (where $t = \frac{1}{x}$), $1 = \lim_{t\to 0} \cos x \le \lim_{t\to 0} \frac{\sin t}{t} \le \lim_{t\to 0} 1 = 1$, thus $\lim_{t\to 0} \frac{\sin t}{t} = \lim_{x\to\infty} \frac{\sin(\frac{1}{x})}{\frac{1}{x}} = 1$

(3) $y = (1+\frac{2}{n})^n$

$\lim_{n\to\infty} (1+\frac{2}{n})^n = \lim_{n\to\infty} (1+\frac{1}{\frac{n}{2}})^n = \lim_{n\to\infty} \left[(1+\frac{1}{\frac{n}{2}})^{\frac{n}{2}}\right]^2 = e^2$

converges

(4) $a_n = \ln(2n^2+1) - \ln(n^2+1) = \ln\left(\frac{2n^2+1}{n^2+1}\right) = \ln(\frac{2+\frac{1}{n^2}}{1+\frac{1}{n^2}}) \to \ln 2$ as $n \to \infty$. converges.

(5) $a_n = n - \sqrt{n+1}\sqrt{n+3} = n - \sqrt{n^2+4n+3} = (n - \sqrt{n^2+4n+3}) \cdot \frac{n+\sqrt{n^2+4n+3}}{n+\sqrt{n^2+4n+3}}$

$= \frac{n^2-(n^2+4n+3)}{n+\sqrt{n^2+4n+3}} = \frac{-4n-3}{n+\sqrt{n^2+4n+3}} = \frac{-(4n+3)/n}{(n+\sqrt{n^2+4n+3})/n} = \frac{-4-\frac{3}{n}}{1+\sqrt{1+\frac{4}{n}+\frac{3}{n^2}}}$

So $\lim_{n\to\infty} a_n = \frac{-4-0}{1+\sqrt{1+0+0}} = -2$ converges.

4. (a) $a_1=1$. $a_{n+1}=4-a_n$ for $n\geq 1$.

$a_1=4$. $a_2=4-a_1=4-1=3$. $a_3=4-a_2=4-3=1$. $a_4=4-a_3=4-1=3$. $a_5=4-a_4=1$.

Since the terms of sequence alternate between 1 and 3. the sequence diverges.

(b) $a_1=2$. $a_2=4-a_1=2$, $a_3=4-a_2=2$. Since all of the terms are 2. So $\lim_{n\to\infty} a_n=2$.

the sequence converges.

5. $\lim_{n\to\infty} a_n=L$. $\lim_{n\to\infty} b_n=M$.

(a) We first show that $\{a_n\}$, $n\geq 1$ is bounded.

For $\varepsilon=1$. we can find that $N>0$ such that $|a_n-L|<\varepsilon=1$.

which implies $|a_n|=|a_n-L+L|\leq 1+L$, $n\geq N$.

So if we set $k=\max\{|a_1|,\cdots|a_{n-1}|,1+L\}$, then $|a_n|\leq k$ for all $n\in\mathbb{N}$.

Let $\varepsilon>0$ be arbitrarily given. Then $\exists N_1, N_2$, such that

$$|a_n-L|<\frac{\varepsilon}{2(k+|M|+1)},\quad n\geq N_1.$$

$$|b_n-M|<\frac{\varepsilon}{2(k+|M|+1)},\quad n\geq N_2.$$

Then for $n\geq N=\max\{N_1,N_2\}$.

$$\begin{aligned}|a_nb_n-LM| &= |a_nb_n-a_nM+a_nM-LM|\\ &\leq |a_n|\cdot|b_n-M|+|M|\cdot|a_n-L|\\ &< k\cdot\frac{\varepsilon}{2(k+|M|+1)}+|M|\cdot\frac{\varepsilon}{2(k+|M|+1)}\\ &<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon.\end{aligned}$$

(b). we only need to show that $\lim_{n\to\infty}\frac{1}{b_n}=\frac{1}{M}$. (*)

since it implies $\lim_{n\to\infty}\frac{a_n}{b_n}=\lim_{n\to\infty}a_n\cdot\frac{1}{b_n}=L\cdot\frac{1}{M}=\frac{L}{M}$.

We now show (*). Let $\varepsilon>0$ be given.

Note that $\left|\frac{1}{b_n}-\frac{1}{M}\right|=\frac{|M-b_n|}{|b_nM|}=\frac{|b_n-M|}{|b_n||M|}$

$\exists N_1$ such that for $n\geq N_1$.

$$|b_n\beta-M|<\frac{|M|}{2}.$$

which implies $|b_n|=|b_n-M+M|\geq|M|-|b_n-M|=\frac{|M|}{2}$. if $n\geq N_1$. (1)

$\exists N_2$ such that for $n \geq N_2$.

$|b_n - M| < \frac{|M|^2 \varepsilon}{2}$ (2).

By (1) and (2), for $n \geq N = \max\{N_1, N_2\}$.

$$\left|\frac{1}{b_n} - \frac{1}{M}\right| = \frac{|b_n - M|}{|b_n||M|} < \frac{\frac{|M|^2}{2}\varepsilon}{\frac{|M|}{2} \cdot |M|} = \varepsilon.$$

Showing (x)

6. if $|r| \geq 1$. then $\{r^n\}$ diverges. So $\{nr^n\}$ diverges also. Since $|nr^n| = n|r^n| \geq |r^n|$.

if $|r| \leq 1$. $\lim_{x\to\infty} x r^x \to \lim_{x\to\infty} \frac{x}{r^{-x}}$

let $k = \frac{1}{|r|}$, $k > 1$.

let $k^n = (1+q)^n = \sum_{i=1}^{n} C_n^i q^i = 1 + nq + q^2 \frac{n(n-1)}{2} + \cdots \Rightarrow q^2 \frac{n(n-1)}{2}$

$$\lim_{x\to\infty}\left|\frac{x}{r^{-x}}\right| = \lim_{x\to\infty} \frac{x}{k^x} \leq \lim_{x\to\infty} \frac{x}{q^2 \frac{x(x-1)}{2}} = \lim_{x\to\infty} \frac{\frac{2}{x}}{q^2(1-\frac{1}{x})} = 0$$

$$0 = -\lim_{x\to\infty} \frac{\frac{2}{x}}{q^2(1-\frac{1}{x})} \leq \lim_{x\to\infty} \frac{x}{r^{-x}} \leq \lim_{x\to\infty} \frac{\frac{2}{x}}{q^2(1+\frac{1}{x})} = 0 \quad \text{. thus } \lim_{x\to\infty} \frac{x}{r^{-x}} = 0.$$

and hence $\{nr^n\}$ converges where $|r| \leq 1$

7. (a) let $\lim_{n\to\infty} a_n = L$. For every $\varepsilon > 0$ there is an integer N such that $|a_n - L| < \varepsilon$ where $n > N$. Thus $|a_{n+1} - L| < \varepsilon$ where $n+1 > N \Leftrightarrow n > N-1$.

It follows that $\lim_{n\to\infty} a_{n+1} = L$ and also $\lim_{n\to\infty} a_n = \lim_{n\to\infty} a_{n+1}$.

(b). If $L = \lim_{n\to\infty} a_n$ then $\lim_{n\to\infty} a_{n+1} = L$ also, so L must satisfy $L = \frac{1}{1+L} \Rightarrow$

$L^2 + L - 1 = 0 \Rightarrow L = \frac{-1+\sqrt{5}}{2}$ (L must be nonnegative)

8. For $\{\sqrt{2}, \sqrt{2\sqrt{2}}, \sqrt{2\sqrt{2\sqrt{2}}}, \cdots\}$, $a_1 = 2^{\frac{1}{2}}$, $a_2 = 2^{\frac{3}{4}}$, $a_3 = 2^{\frac{7}{8}}$, $\cdots$. So

$a_n = 2^{\frac{2^n - 1}{2^n}} = 2^{1-(\frac{1}{2})^n}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} 2^{1-(\frac{1}{2})^n} = 2^1 = 2.$

9. $a_1 = 1$. $a_{n+1} = 3 - \frac{1}{a_n}$

By induction than $\{a_n\}$ is increasing and bounded above by 3.

Let P_n be the proposition that $a_{n+1} > a_n$ and $0 < a_n < 3$. Clearly P_1 is true.

Assume that P_n is true. Then $a_{n+1} > a_n \Rightarrow \frac{1}{a_{n+1}} < \frac{1}{a_n} \Rightarrow -\frac{1}{a_{n+1}} > -\frac{1}{a_n}$.

Now $a_{n+2} = 3 - \frac{1}{a_{n+1}} > 3 - \frac{1}{a_n} = a_{n+1} \Leftrightarrow P_{n+1}$.

This proves that $\{a_n\}$ is increasing and bounded above by 3, so $1 = a_1 < a_n < 3$.

that is, $\{a_n\}$ is bounded, and hence convergent.

If $L = \lim_{n\to\infty} a_n$, then $\lim_{n\to\infty} a_{n+1} = L$ also, so L must satisfy $L = 3 - \frac{1}{L} \Rightarrow$

$L^2 - 3L + 1 = 0 \Rightarrow L = \frac{3 \pm \sqrt{5}}{2}$, but $L > 1$, so $L = \frac{3+\sqrt{5}}{2}$.

Bonus.

1. $\forall \varepsilon > 0$, $\exists N > 0$, such that if $m \geq N$ and $n \geq N$, $|x_m - x_n| < \varepsilon$.

By triangle inequality, for $n, m \geq N$,

$|\,|x_m| - |x_n|\,| \leq |x_m - x_n| < \varepsilon$, so $\{|x_n|\}$ is a Cauchy sequence.

2. $|q| < 1$, $|a_k| \leq M$ for $k = 1, 2, \cdots$

$\forall \varepsilon > 0$, $\exists N > 0$. such that if $m > N$ and $n > N$, $|a_m - a_n| < \varepsilon$.

Suppose $m > n$.

$$|x_m - x_n| = |a_0 + a_1 q + a_2 q^2 + \cdots + a_m q^m - (a_0 + a_1 q + \cdots + a_n q^n)|$$

$$= |a_{n+1} q^{n+1} + a_{n+2} q^{n+2} + \cdots + a_m q^m)$$

$$\leq M \cdot (|q^{n+1}| + |q^{n+2}| + |q^m|) \quad (\text{set } |q| = r)$$

$$= M r^{n+1} (1 + r + \cdots + r^{m-n-1})$$

$$= M r^{n+1} \frac{1 \cdot (1 - r^{m-n})}{1 - r}$$

$$\leq \frac{M r^{n+1}}{1 - r}$$

Note that $\frac{M r^{n+1}}{1-r} < \varepsilon \Leftrightarrow n > \log_r\left(\frac{\varepsilon(1-r)}{M}\right) - 1$

we get $N = \left[\log_r\left(\frac{\varepsilon(1-r)}{M}\right) - 1\right] + 1$

$$= \left[\log_r\left(\frac{\varepsilon(1-r)}{M}\right)\right]$$